

sec^{-1} erhalten [17]. Die Werte für A_∞ in Tabelle 2 zeigen, daß einfache Abschätzungen des Vorfaktors unter Anwendung der Theorie des Übergangszustandes (vgl. [4]) bei diesem Reaktionstyp, zumindest in einigen Fällen, nicht zu befriedigender Übereinstimmung mit den experimentellen Ergebnissen führen.

Die Vorfaktoren der Arrhenius-Ausdrücke für die beiden untersuchten Zerfallsreaktionen im Niederdruckbereich wurden nach Gleichung (4) bestimmt. Bei der Berechnung der gaskinetischen Stoßzahl bei 1650 K wurden die folgenden Stoßdurchmesser benutzt: 3,1 Å für Ar; 6,2 Å für C_2Cl_4 ; 5,8 Å für C_2HCl_3 . Für die Zustandsdichte $\rho_{\text{vib}}(E_0)$ der Schwingungszustände an der Dissoziationsgrenze E_0 wurden nach einem quantenstatistischen Ausdruck von Thiele [18] in harmonischer Näherung die Werte $4,8 \cdot 10^{12} \text{ mol kcal}^{-1}$ für C_2Cl_4 und $1,7 \cdot 10^{11} \text{ mol kcal}^{-1}$ für C_2HCl_3 erhalten. Für den Korrekturfaktor $A(T)$, dessen Größe u. a. von Einzelheiten der Energieübertragung in Stößen und von Einflüssen der Molekülrotation bestimmt wird, sind aus Ergebnissen über den thermischen Zerfall vieler drei- und vieratomiger Moleküle für Argon als Stoßpartner Werte zwischen 0,1 und 1 kcal/mol erhalten worden (vgl. [11]). Bei vergleichbaren und etwas größeren Molekülen als den hier beschriebenen Äthylenverbindungen liegen z. Zt. erst wenige experimentelle Daten zum Vergleich vor, da sich der Niederdruckbereich bei ihnen experimentell oft nicht mehr erreichen läßt und eine zuverlässige Extrapolation der gemessenen Geschwindigkeitskonstanten zu niedrigen Drücken erforderlich wird. Durch Vergleich der nach Gleichung (4) berechneten und der experimentell erhaltenen Geschwindigkeitskonstanten für den Niederdruckbereich erhält man $A(T) = 0,5 \text{ kcal/mol}$ für C_2Cl_4 und $A(T) = 0,4 \text{ kcal/mol}$ für C_2HCl_3 , jeweils bei 1650 K (vgl. Tabelle 2). Diese Werte für $A(T)$ liegen ebenso wie diejenigen für die Isomerisierung von Methylisocyanid ([19], vgl. [11]) sowie für den thermischen Zerfall von Äthylen [22], Nitromethan [20] und Nitroäthan [21] in dem oben aufgeführten, für kleinere Moleküle gefundenen Bereich.

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Specific Rate Constants of Unimolecular Processes II. Adiabatic Channel Model

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Reaktionskinetik

The rate of unimolecular processes is calculated by means of a statistical adiabatic channel model. The rates are mainly determined by the maxima of the channel energies. The channels are constructed by correlating reactant and product states; the channel energies are computed by a simple interpolation procedure. The coupling between the various vibrational-rotational motions is taken into account. The influence of parameters of the potential surface on channel energies and on rate constants is studied. Numerical results for the NO_2 , CH_4 , CD_4 , C_2H_6 and C_2D_6 dissociation-recombination kinetics are compared to experimental data. The statistical adiabatic channel model gives similar results as minimum local entropy models. Both these approaches lead to stronger curvatures of $k(E)$ and smaller activation energies for the thermal rate constant k_∞ than the corresponding RRKM calculations.

Die Geschwindigkeit unimolekularer Zerfallsreaktionen wird mit Hilfe eines Modells adiabatischer Kanäle statistisch berechnet. Wesentlich für die Reaktionsgeschwindigkeiten sind die Maximalenergien der Kanäle, die durch Korrelation von Zuständen der Reaktanten und Produkte und einfache Interpolation bestimmt werden. Die Kopplung der Schwingungs-Rotationsbewegungen wird beachtet. Der Einfluß von Potentialflächenparametern auf Kanalenergien und Geschwindigkeitskonstanten wird studiert. Numerische Ergebnisse für die Dissoziations-Rekombinationskinetik von NO_2 , CH_4 , CD_4 , C_2H_6 und C_2D_6 werden mit experimentellen Daten verglichen. Das Modell adiabatischer Kanäle ergibt ähnliche Ergebnisse wie Modelle mit aktivierten Komplexen, die an Stellen minimaler Zustandsdichte lokalisiert werden. Beide Ansätze führen zu einer stärkeren Krümmung der $k(E)$ -Kurven und zu geringeren Arrhenius-Aktivierungsenergien für die thermischen Geschwindigkeitskonstanten k_∞ als entsprechende RRKM Rechnungen.

I. Introduction

The RRKM [1, 2] and the Quasi-equilibrium [3, 4] versions of the transition state theory have successfully been applied to many experimental studies of unimolecular processes [5–7]. The characterization of the activated complex is the central problem of these applications. With a "good choice" one can often fit the experimental data quite well and compensate for basic imperfections of the theory. Though useful in practice, this procedure becomes problematic if one wants to evaluate kinetic data in terms of otherwise inaccessible properties of the potential surface. Therefore, on the one hand one should try to improve the theoretical model, as far as this is possible without losing its great versatility in experimental applications. On the other hand one should use potential surfaces which describe the transition between the known reactant and product configurations by one adjustable parameter only. If a description of the experiments would really require more than one parameter, one should reinspect the model before further potential parameters are allowed for.

To have a qualitatively correct picture of the potential surface the following points have to be observed. The kinetics is particularly sensitive to the poorly known intermediate range between reactant and product configurations. Here, the eigenvalues in the coordinate of the reaction path and the other coordinates cannot be separated. In bond fission reactions some of the bending vibrations of the reactant become at first hindered, then free rotations of the fragments relative to each other. A harmonic oscillator approximation for these coordinates is certainly inadequate. The coupling between all of the rotational motions, including these hindered rotations, has to be taken into account properly. The reaction can be characterized by a smooth transition along the reaction path of the reactant eigenvalues to the product eigenvalues. In order to determine activated complex configurations one has to derive this transition from the shape of the potential surface. In part I of this series [8] this was done using the bond energy-bond order method [9] to account for the dependence of force constants on bond lengths, and a local normal coordinate analysis at each value of the reaction coordinate. The corresponding potential parameter initially was identified with that of stretching force constant-bond length correlations [10]. In the present work this parameter is used instead as the only adjustable parameter of the potential surface. The transition bending vibration $\rightarrow$ hindered rotation $\rightarrow$ free rotation can be described by a new technique, proposed in the present paper, which avoids a local normal coordinate analysis along the reaction path. One would hope that quantum-chemical calculations in the future will provide the corresponding potentials

of these particularly important hindered rotations as a function of the bond length.

To locate the position of "the activated complex" is a particularly important question in simple bond fission reactions with Morse-type potential surfaces. Here, there is no pronounced barrier in the electronic potential between reactant and products which would correspond to the intuitive picture of "the activated complex". The "minimum local density of states" criterion of Bunker and Pattengill [11] provided a way out of this dilemma. This criterion identifies the minimum of some local density of states or some local entropy with the dynamic bottle-neck of the reaction. These "activated complexes" generally do not coincide with the centrifugal barriers of the rotating molecules. Using this criterion, at first, activated complexes for average reacting molecules in thermal reactions have been determined [11, 12]. The equivalence with the old maximum Gibbs energy criterion for canonical ensembles has been demonstrated [13]. The extension to microcanonical ensembles leads to energy dependent positions of activated complexes in one reaction [7, 8]. As the criterion is intuitive, it can be formulated in different ways with differing numerical results. Therefore, one should try to "prove" it by more extended forms of the theory. Indeed, the criterion in its original form agrees well with classical trajectory calculations for triatomic model systems [11]. Larger numerical differences between the various possible ways of applying the criterion are expected for larger polyatomic systems. For this reason we propose in the present work a different model which offers some conceptual advantages, because it is derived from resonance scattering theory [14–16]. In this formalism localized activated complex configurations can be replaced by "open reaction channels" correlating reactant and product states. Explicite quantum scattering calculations have rarely been performed yet. Nevertheless, the use of a statistical postulate can bring such a model to the working level of a RRKM-formalism. Corrections for special quantum effects or non-statistical situations can be introduced in a most natural way whenever necessary. The present model, therefore, can be used to test the different versions of the minimum local entropy criterion, or to replace them in many cases because it is not much more difficult to handle.

II. Statistical Model of Open Adiabatic Reaction Channels

We shall consider unimolecular processes, taking place on one potential surface, e.g. spin allowed dissociation or isomerization reactions of polyatomic molecules. These reactions can be described by the formalism of resonance scattering theory or theories of radiationless transitions [15–19].

Although there have been early attempts in this direction [20, 21], detailed formulations have been worked out only recently. In the following we choose the approach of Mies and Kraus [15, 16] (see also [22]), which provides a language easily adaptable to that of the transition state theory.

The resonance scattering states $|n\rangle$ can be approximated by zero order rovibrational states $|a_n\rangle |\lambda_n\rangle$ of the reactant molecule with bound states in the reaction coordinate ($|\lambda_n\rangle$) and in the other coordinates ($|a_n\rangle$). For dissociations, the product states can be approximated by $|p\rangle \simeq |a_p\rangle |e_p\rangle$ with continuum states $|e_p\rangle$ in the reaction coordinate. (For isomerizations both spectra contain only bound states). Following the usual representation of experimental data, we are interested in specific rate constants $k_{n \rightarrow p}(E)$, i.e. in the first order rate constants for the unimolecular decay of $|n\rangle$ to $|p\rangle$ at a total rovibrational energy between E and $E + dE$. As shown in [15, 16], $k_{n \rightarrow p}(E)$ is given by expressions of the form

$$k_{n \rightarrow p}(E) = \frac{\gamma_{n \rightarrow p}(E)}{h \cdot \rho_p(E)} \quad (1)$$

where $\rho_p(E)$ is the density of those states $|n\rangle$, which are coupled to the particular product state $|p\rangle$ (on the time scale of the experiment). Expressions for the "transmission coefficient" $\gamma_{n \rightarrow p}(E)$ e.g. have been derived in [15] (for a model of resonances with equal spacings and widths, observing the effect of the overlapping widths of the resonance states $|n\rangle$). $\gamma_{n \rightarrow p}$ is limited by $0 \leq \gamma_{n \rightarrow p} \leq 1$. If $\rho_p(E)$ is identified with the total density $\rho(E)$ of the states $|n\rangle$, the limit for $\gamma_{n \rightarrow p}(E)$ imposes an upper bound on $k_{n \rightarrow p}(E)$ only if all states $|n\rangle$ couple equally well with $|p\rangle$. Combining all product channels $|a_p\rangle$ accessible at the energy E with total angular momentum J , one obtains the total specific rate constant:

$$k(E, J) = \sum_{a_p} k_{n \rightarrow p}(E) = \sum_{a_p} \frac{\gamma_{n \rightarrow p}(E, J)}{h \rho_p(E, J)} \quad (2)$$

(More generally J stands for all good quantum numbers that are conserved during reaction).

The explicit calculation of the transmission coefficient $\gamma_{n \rightarrow p}$ and the specification of those states $|n\rangle$ that couple with $|p\rangle$ presents a most difficult task. In order to arrive nevertheless at a simple version of the theory, which can be handled easily enough for experimental applications, one uses statistical postulates analogous to transition state theory:

i) one assumes that all states $|n\rangle$ in the range $(E, E + dE)$ are equally well coupled with the accessible product states $|p\rangle$ such that:

$$\rho_p(E, J) \simeq \rho(E, J). \quad (3)$$

If there is information on restricted coupling, this could be introduced as a "nonstatistical modification" at this point.

ii) One assumes equal statistical weight for all accessible product channels $|a_p\rangle$. To determine the accessible channels, one might construct a suitable set of curvilinear coordinates (R_q, q) as described in [16, 23, 24] (q = reaction coordinate, R_q = other coordinates). Then, from the Hamiltonian $H(R_q, q, \frac{\partial}{\partial q})$ one obtains a set of "adiabatic channel states", $|a_q\rangle$, which go continuously from reactant states

$|a_n\rangle$ to product states $|a_p\rangle$ with eigenvalues $E_a(q)$. The sum of the eigenvalues $E_a(q)$ and the potential energy $V(q)$ in the reaction coordinate shall be denoted as "adiabatic potential curves $V_a(q)$ ". Some of them are illustrated in Fig. 1. At a

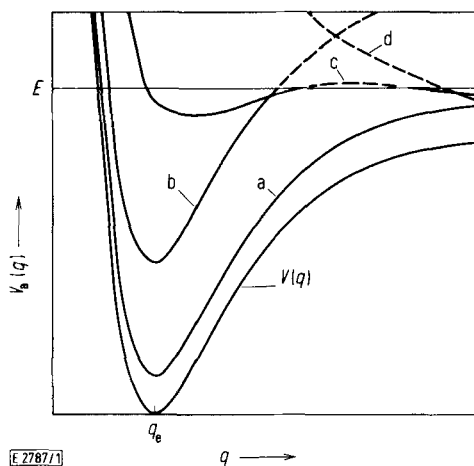


Fig. 1
Adiabatic potential curves $V_a(q) = V(q) + E_a(q)$ (a = open, b – d = closed channels)

given total energy E , the regions under the dotted parts of these curves are adiabatically not accessible. (The sense of this adiabatic approximation becomes clearer when a dynamic model, as given below, is considered.) The adiabatic approximation can be written

$$\gamma_{n \rightarrow p}(E) \simeq 0 \quad \text{if} \quad E < V_{a \max} \quad (4)$$

which is complemented by the statistical postulate ii)

$$\gamma_{n \rightarrow p}(E) \simeq 1 \quad \text{if} \quad E > V_{a \max} \quad (5)$$

$V_{a \max}$ is the maximum of the adiabatic potential curves for $q_e < q < \infty$. q_e is the value of the reaction coordinate in the equilibrium position of the reactant. With the assumption (4) one excludes diabatic quasitunneling through barriers of the type c) of Fig. 1 via jumping to another adiabatic channel state. Furthermore, ordinary tunneling through centrifugal and other barriers is neglected. The statistical postulates i) and ii) together with the adiabatic approximation (4) will be used throughout this work. In principle, these assumptions could be relaxed.

With these approximations the specific rate constant is given by

$$k(E, J) \simeq \frac{\sum_{a_p} 1}{h \rho(E, J)} \equiv \frac{W(E, J)}{h \rho(E, J)} \quad (6)$$

$W(E, J)$ is defined in Equation (6) as the sum $\sum_{a_p}$ of adiabatic reaction channels that are open according to Equations (4) and (5). In the present model there are no localized activated complexes, whose number of quantum states would be given by $W(E, J)$. Instead, $W(E, J)$ is determined by the whole set of the maxima of adiabatic potential curves, which are located between $q = q_e$ and $q = \infty$, and with $V_{a \max} < E$.

Before constructing the adiabatic potential curves $V_a(q)$, it seems worthwhile to illustrate Equation (1) by a model used since long in nuclear physics [14]. Suppose that the excitation process has produced some linear superposition of the states $|n\rangle$. The recurrence time for a particular dynamic state of the system is

$$t_r = h \cdot \rho(E, J). \quad (7)$$

This time may be identified with the average time to form from an arbitrary activated state a critical configuration, which is described by an outgoing wave in the reaction coordinate and quantum numbers a_q for the other coordinates. It is supposed that the system stays adiabatically in $|a_q\rangle$ during the spread out of the wave. The wave is then transmitted with a probability T_a at the adiabatic potentials in Fig. 1. Multiplying with the number of approaches per unit time, one obtains the specific rate constant for one channel

$$k_a(E) \simeq \frac{T_a}{h \cdot \rho(E, J)}. \quad (8)$$

The general threshold behaviour for barriers of the considered type is well known [25]

$$T_a(E) \propto \sqrt{E - V_{a\max}} \quad \text{at} \quad E \geq V_{a\max}. \quad (9)$$

In order to estimate the energy range in which $T_a(E)$ rises from zero to values near 1 one can consider a truncated Morse potential [26]

$$V(a) = V_{\max} [1 + \exp\{-2(q - q_{1/2})/L\}]^{-1}. \quad (10)$$

The steepness parameter $L \simeq 1.2\beta^{-1}$ gives the same slope as a full Morse potential (Morse parameter β) at half the barrier height $q_{1/2}$. For this potential,

$$T(E) = 1 - \left\{ \frac{\sinh \pi L(k - K)/2}{\sinh \pi L(k + K)/2} \right\}^2 \quad (11)$$

with $k \simeq \sqrt{2mE/\hbar^2}$ and $K \simeq \sqrt{2m(E - V_{\max})/\hbar^2}$. For typical molecular cases the rise of T_a is quite steep ($T \simeq 0.9$ at $E \simeq (1 + 6 \cdot 10^{-5}) \cdot \Delta H_0^0$ for NO_2 dissociation with $\Delta H_0^0 = 72$ kcal/mol and $\beta = 3.2 \text{ \AA}^{-1}$). This effect, which could easily be introduced in Equations (1) and (2), might contribute to "meta-stable" decompositions near the threshold.

III. Construction of Adiabatic Potential Curves

In this section we shall be concerned with an approximate determination of the adiabatic potential curves

$$V_a(q) = V(q) + E_a(q) \quad (12)$$

which are required to determine the number of open channels $W(E, J)$ in Equation (6).

Channel eigenvalues $E_a(q)$: In order to construct the eigenvalues $E_a(q)$, one first has to give a correlation between reactant states $|a_n\rangle$, adiabatic channel states $|a_q\rangle$ and product states $|a_p\rangle$. Two components in $|a_q\rangle$ will be treated in different ways:

i) Some quantum numbers can be considered as being conserved individually during the reaction. First, there are the quantum numbers J and M of the molecular angular momentum (excluding electronic and nuclear spin contributions). Further, one can normally find a fairly unique one-to-one

correlation between the oscillators of the fragments (quantum numbers v_{ip} with $i = 1, \dots, r$) and some reactant oscillators (quantum numbers v_{in} with $i = 1, \dots, r$). For all of these we assume: $v_{in} = v_{ip} = v_{iq} \equiv v_i$.

ii) To specify completely the states $|a_n\rangle$ of the reactant one has to add the quantum numbers v_{bn} for the remaining, mainly bending vibrations, and the rotational quantum number τ (for an asymmetric top) or K (for a symmetric top).

The states of the separated fragments $|a_p\rangle$ are characterized by their rotational quantum numbers $j_1, m_1, \tau_1, j_2, m_2, \tau_2$ (assuming two asymmetric tops) and the quantum numbers l and m_l for the orbital motion of the two fragments. There is no way of defining any of these quantum numbers individually in the intermediate range, because of the strong coupling of the corresponding motions. Therefore, these components of the states $|a_q\rangle$ are characterized by a single quantum number w , which labels reactant and product states in the order of increasing energy and of increasing l for degenerate product levels.

For this one-to-one correlation of reactant and product states, one has to observe, of course, eventual symmetry restrictions (see below) and the conditions

$$|\tau| \leq J; \quad |\tau_1| \leq j_1; \quad |\tau_2| \leq j_2;$$

$$|m_1| \leq j_1; \quad |m_2| \leq j_2; \quad |M| \leq J;$$

$$\vec{j}_1 + \vec{j}_2 + \vec{l} = \vec{J} \quad \text{and} \quad m_1 + m_2 + m_l = M.$$

Symbolically, one may describe the construction of the adiabatic channel states $|a_q\rangle$ by a correlation diagram:

$$\begin{array}{ccccccc} J & & \dots & J & \dots & J & \\ M & & \dots & M & \dots & M & \\ v_{1n} & & \dots & v_1 & \dots & v_{1p} & \\ v_{2n} & & \dots & v_2 & \dots & v_{2p} & \\ \vdots & & & \vdots & & \vdots & \\ v_{rn} & & \dots & v_r & \dots & v_{rp} & \\ (v_{bn}, \tau_n) & \dots & w & \dots & (j_1, m_1, \tau_1, j_2, m_2, \tau_2, l, m_l) & & \\ \hline |a_n\rangle & & \dots & |a_q\rangle & \dots & |a_p\rangle & \end{array}$$

with $v_{in} = v_{ip} = v_i$.

In the absence of any precise knowledge on the potential surface in the intermediate range, a simple interpolation between the limiting values is used to obtain the eigenvalues of $|a_q\rangle$. The interfragment distance is used as reaction coordinate q . One oscillator of the reactant molecule is identified with the motion in this coordinate. The interpolation formula for the eigenvalues in the other coordinates is:

$$\begin{aligned} E_a(q) = & E_a(q_e, v_{in}, v_{bn}, \tau) \cdot \exp[-\alpha(q - q_e)] \\ & + E_a(q = \infty, v_{ip}, j_1, j_2, \tau_1, \tau_2) \cdot (1 - \exp[-\alpha(q - q_e)]) \\ & + E_{\text{cent}}. \end{aligned} \quad (13)$$

The centrifugal energy E_{cent} is calculated separately (see Equations (14), (15)). The exponential interpolation with a parameter α corresponds to empirical relations such as Badger's rule, Pauling's rule and the BEBO method [9] (see below). q_e denotes the equilibrium value of q in the reactant.

For illustration, we shall consider explicitly the dissociation of a bent triatomic molecule, which is almost a symmetric

top. In this case, $|a_p\rangle$ is described by the quantum number v_p of the vibration of the diatomic fragment, its rotational quantum number j and the orbital quantum number l for the relative motion of the two fragments. Rotational quantum numbers J and M are the same for both $|a_p\rangle$ and $|a_n\rangle$. $|a_n\rangle$ is specified with the quantum numbers v_n for one stretching, v_{bn} for the bending vibration and the symmetric top quantum number $K \leq J$. The coupling of angular momenta leads to

$$|J - j| \leq l \leq J + j.$$

If the diatomic is heteronuclear, there are no important symmetry restrictions. The adiabatic correlation is then

$$(J, M, v_n, [v_{bn}, K]) \rightarrow (J, M, v_p = v_n, [j, l])$$

with the eigenvalues of $|a_q\rangle$:

$$E_a(q) \approx [(v_n + \frac{1}{2})\varepsilon_n + (v_{bn} + \frac{1}{2})\varepsilon_{bn} + \{A_n(q_e) - B_n(q_e)\} \cdot K^2] \exp(-\alpha(q - q_e)) + [(v_p + \frac{1}{2})\varepsilon_p + A_p j(j + 1)](1 - \exp(-\alpha(q - q_e))) + B_s(q) \cdot P(P + 1) \quad (14)$$

with

$$P = J \cdot \exp[-\alpha(q - q_e)] + l \cdot \{1 - \exp[-\alpha(q - q_e)]\} \quad (15)$$

ε_n and ε_{bn} are the stretching and bending quanta of the reactant molecule. $A_n(q_e)$ and $B_n(q_e) = \frac{B(q_e) + C(q_e)}{2}$ are its rotational constants. ε_p is the stretching quantum, A_p the rotational constant of the diatomic fragment. $B_s(q) = \frac{B(q) + C(q)}{2}$ is the mean of the small rotational constants which we calculated from the inertial tensor for a fixed bending angle.

The centrifugal term with the pseudo-quantum number P from Equation (15) represents correctly rotational energies for small and relatively large bond extensions $(q - q_e)$, and it gives a simple interpolation between these limiting values for intermediate values of q .

The eigenvalues $E_a(q)$ and the correlation scheme are illustrated in Fig. 2 for some combinations of quantum numbers for dissociating NO_2 with $J = 10$ and $v_n = v_p = v_q = 0$.

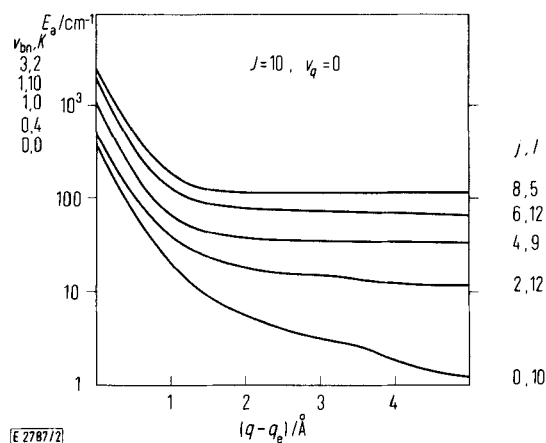


Fig. 2

Correlations and eigenvalues $E_a(q)$ for some channels of a bent triatomic molecule, which is almost a symmetric top. The energy scale applies approximately to NO_2 , with $J = 10$ and $v_{\text{stretch}} = 0$

Potential energy in the reaction coordinate $V(q)$: For simple bond fission reactions, the potential $V(q)$ in Equation (12) can be approximated by a Morse potential

$$V(q) = D \{1 - \exp[-\beta(q - q_e)]\}^2 \quad (16)$$

with

$$D = \Delta H_0^0 + E_{zn} - E_{zp} \quad (17)$$

where ΔH_0^0 = bond dissociation energy, E_{zn} = zero point energy of the reactant, E_{zp} = zero point energy of the fragments. The Morse parameter β is determined from the force constant F_q for the dissociating bond,

$$\beta = \sqrt{\frac{F_q}{2D}} \quad (18)$$

of from the molecular frequencies, see [27]. Since the Morse curve is notoriously too flat for large q , one might use an increased β in this range (for instance the known potential curve of H_2 [28] can be represented by a Morse potential with β increasing by about a factor of 1.5 at $q - q_e \rightarrow 3 \text{ \AA}$. Explicite calculations of $V_a(q)$ show that the use of a Morse potential starting from reactant properties generally is more useful than a van der Waals potential starting at the product side.

Pauling parameter α : The "Pauling parameter" α in Equations (13), (14) is less directly derived from known data than the Morse parameter β . α will therefore be treated as adjustable parameter. Nevertheless, its magnitude can be estimated. As shown in [9, 10], stretching force constants F of diatomic molecules are approximately related by

$$F \approx F_s \cdot n. \quad (19)$$

F_s is the force constant of a single bond. The bond order n is defined by Pauling's relation

$$q_e - q_{es} \equiv -(0.26 \text{ \AA}) \ln n \quad (20)$$

with the equilibrium bond length q_{es} for a single bond. By analogy, without further empirical justification, a relation

$$F_\phi = F_{\phi s} \cdot n_1 \cdot n_2 \quad (21)$$

has been proposed [9] for bending force constants F_ϕ , where n_1 and n_2 are the bond orders of the bonds adjacent to the bending angle. As $\varepsilon \propto \sqrt{F}$ at sufficiently large q , this would correspond to $\alpha \approx 1.9 \text{ \AA}^{-1}$ in Equations (13), (14). However, when we tried to find Badger type relations for bending force constants, we could establish them only for groups of molecules with similar bond angles, leading then rather to $\alpha \approx 0.5 \text{ \AA}^{-1}$. Similar tendencies leading to low α -values near $1 \text{ \AA}^{-1}$ are observed when plotting bending force constants versus some excess bond length ("excess" with respect to standard covalent bond radii etc.). Evaluation of the potential surface of LiH_2^+ , which is quite well known from quantum-chemical calculations [29], leads again to similar values of α . The range of "reasonable" values for α is then rather limited.

IV. Densities of States, Symmetry and Reaction Path Degeneracy

The density of states $\rho(E, J)$ is defined as the number of quantum states per unit energy with some fixed J and a total

rovibrational energy E . For a symmetric top this can be evaluated with

$$\rho(E, J) = \sum_{K=0}^J g_K \cdot \rho_{\text{vib}} [E - J \cdot (J+1) \cdot B - (A-B) \cdot K^2 - E_z] \quad (22)$$

where $g_K = 1$ for $K = 0$ and $g_K = 2$ otherwise. ρ_{vib} can be estimated with the semiclassical formula for harmonic oscillators:

$$\rho_{\text{vib}}(E_{\text{vib}}) = \frac{(E_{\text{vib}} + E_z)^{s-1}}{(s-1)! \prod_{i=1}^s \varepsilon_i} \quad (23)$$

where the ε_i are the vibrational quanta; the formulae actually used in our calculations are those given in [30] for anharmonic vibrational densities of states, which we have also checked by direct count. For low J and small $(A-B)$ one has roughly $\rho(E, J) < (2J+1)\rho_{\text{vib}}[E - J(J+1) \cdot B - E_z]$. Computing $\rho(E, J)$ takes account of the fact that states with different J are not coupled. Furthermore Equation (22) implicitly assumes that also states with different nuclear spin functions are not coupled, i.e. the nuclear spin function is supposed to be conserved during the dissociation process. It might be that there are further quantum numbers v_a which are approximately conserved. In that case $\rho(E, J)$ is replaced by $\rho(E, J, v_a)$. This would lead to a lifetime shortening. For instance in RRKM calculations the rotational quantum number K (or τ) is usually supposed to be conserved with a use of $\rho_{\text{vib}}(E_{\text{vib}})$ rather than $\rho(E, J)$. A look at rovibrational perturbations and at the dynamics of quasilinear molecules [27, 31] does, however, not seem to justify a qualitatively different treatment for rotation and vibration (as long as J is conserved). A quantitative decision on this point can be given by quantum mechanical model calculations, classical trajectory calculations or experimental evidence.

If the molecule has some symmetry, not all of the states included in $\rho(E, J)$, really occur for a specified nuclear spin function. On the average, the density calculated in Equation (22) (hereafter ρ_m), which applies to a "mechanical" model of point masses neglecting the possible identity of nuclei, has to be divided by the symmetry number σ of the molecule,

$$\rho(E, J) = \frac{\rho_m(E, J)}{\sigma} \quad (24)$$

Similarly, if the fragments have some symmetry (e.g. symmetry numbers σ_1 and σ_2), only part of the "mechanically" calculated channels occur, and on the average one has

$$W(E, J) = \frac{W_m(E, J)}{\sigma_1 \cdot \sigma_2} \quad (25)$$

Equations (24) and particularly (25) are reasonable approximations only for sufficiently large energy ranges. In order to get correct values in the threshold region of the reaction, one has to investigate molecule and product states according to their transformation properties in the symmetry group of the supermolecule, and to correlate states of the same species. For these channels one then has to compute the nuclear spin statistical weight individually.

In order to illustrate the influence of symmetry we shall briefly consider the dissociation of NO_2 and CH_4 . For N^{16}O_2

the "mechanically" computed rovibrational states of the electronic ground state transform as A_1, A_2, B_1, B_2 in C_{2v} (or in the isomorphous permutation inversion group of the elements (12), (12)*, E^* , E , see [32]). Now, only A_1 and A_2 levels occur, e.g. for $J = 0$ vibrational states with odd quantum numbers in the antisymmetric stretching vibration are missing. At sufficiently high energies $\rho(E, J)$ approaches $\frac{1}{2}\rho_m(E, J)$ as indicated in Equation (24). The term "reaction path degeneracy" for this effect is rather misleading. In $^{16}\text{ON}^{18}\text{O}$, $\rho = \rho_m$ which gives a smaller threshold rate constant $k = (h\rho_m)^{-1}$. However, there are twice as many exit channels corresponding to N^{16}O and N^{18}O . The overall dissociation rate is then approximately the same as for N^{16}O_2 , if one neglects the slight changes in the molecular frequencies. For CH_4 these considerations are much the same, if they are applied to a specified nuclear spin function. If all inversional sublevels are counted [33], one has $\rho = \frac{1}{24}\rho_m$. On the average only 1/6 of the rovibrational levels of CH_3 occur, giving an average "reaction path degeneracy" of $\sigma/(\sigma_1 \cdot \sigma_2) = 4$ for the process $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$. This obviously does not apply in the threshold range. For instance, even without considering the changes of molecular frequencies, the threshold rate constant for CH_2D_2 dissociation is only 1/6 of that for CH_4 (for the one-dimensional symmetry species).

V. Specific and Averaged Rate Constants

In the preceding sections an approximate method has been described to calculate the specific rate constants $k(E, J)$ under the statistical and the adiabatic assumptions. In general, a really specific excitation (e.g. optical selection of a single rovibronic state) will not produce random populations in all other quantum numbers. On the other hand, if one uses non selective broad band activation processes (thermal, optical, electron impact, chemical, complex formation in collision etc.) one has to give the corresponding averages over E and J to obtain the overall rate constants for the process. For instance, in optical excitation one often will produce a distorted thermal distribution of rotational states in the excited state of the reactant, but care has to be taken for overlapping bands. In any case, a thorough analysis of the reactant states involved in a certain experiment has to precede an application of a statistical theory. In the high pressure limit of thermal unimolecular reactions, one has a Boltzmann distribution for the activated molecules. Therefore, after thermal averaging over $k(E, J)$,

$$k_\infty = \frac{kT}{h} \cdot \frac{Q^*}{Q} \exp\left(-\frac{\Delta H_0^0}{kT}\right) \quad (26)$$

with

$$Q^* = \sum_{J=0}^{\infty} (2J+1) \int_0^{\infty} W(E, J) \exp\left(-\frac{E - V_{0\infty}}{kT}\right) \frac{dE}{kT} \\ = \sum_{a_p} \exp\left(-\frac{V_{a\max} - V_{0\infty}}{kT}\right) \quad (27)$$

and the lowest product energy level $V_{0\infty}$.

The sum over the channels a_p is meant to count degenerate channels individually. The "partition function of the activated complex", Q^* , does not correspond to any particular activated

complex configurations with rigid rotor – harmonic oscillator energy levels. Q is the rovibrational partition function for the reactant molecule counted from the lowest eigenstate of the reactant. In terms of “mechanically” (i.e. without symmetry considerations) calculated quantities one has

$$k_{\infty} \simeq \frac{kT}{h} \frac{\sigma}{\sigma_1 \sigma_2} \frac{Q_m^*}{Q_m} \exp\left(-\frac{\Delta H_0^{\ddagger}}{kT}\right) \quad (28)$$

with the reactant symmetry number σ and the fragment symmetry numbers σ_1 and σ_2 .

For large molecules the direct numerical evaluation of $W(E, J)$ becomes very time consuming. One notes, however, that often only few coordinates have eigenvalues that change markedly during extension of the breaking bond. If the number of open channels $W_i(E, J)$ is determined by the described procedure only for these “loose coordinates”, the total $W(E, J)$ can be obtained by convolution:

$$W(E, J) = \int_0^E \rho_r(E - X, J) W_i(X, J) dX. \quad (29)$$

To calculate the density of states $\rho_r(E, J)$ in the other “rigid coordinates” one employs the usual approximations [30, 34].

RRKM Formulae: Later on, we shall compare the present model with the corresponding RRKM calculations. In the RRKM formulation, the specific rate constant would be

$$k(E, J) = n^{\ddagger} \frac{W_m^{\ddagger}(E_{\text{vib}}^{\ddagger})}{h \cdot \rho_m(E_{\text{vib}})} \quad (30)$$

The vibrational energies E_{vib} and $E_{\text{vib}}^{\ddagger}$ depend on the rotational state J . $n^{\ddagger}$ is the reaction path degeneracy [1, 2, 35]. $W^{\ddagger}$ is the number of quantum states of the vibrations and “active” rotations of the activated complex. With a Boltzmann distribution for the activated reactant states one obtains

$$k_{\infty} = n^{\ddagger} \frac{kT}{h} \frac{Q_m^{\ddagger}}{Q_m} \exp\left(-\frac{\Delta H_0^{\ddagger}}{kT}\right). \quad (31)$$

$Q_m^{\ddagger}$ is here the partition function of the localized activated complex in contrast to Equation (27). One assumes [6, 36] that for simple bond fission reactions the activated complexes can be localized at the tops of the centrifugal barriers of the rotating molecule. Also, in the usual [2, 5, 36] treatment rotations and vibrations are separated taking the same vibrational frequencies for the activated complex at all centrifugal maxima. This leads to the factorization $Q_m^{\ddagger} = (Q_{\text{vib}}^{\ddagger} \cdot Q_{\text{rot}}^{\ddagger})_m$. In [37] the coupling of $Q_{\text{rot}}^{\ddagger}$ and $Q_{\text{vib}}^{\ddagger}$ has been considered. Approximate formulae for $Q_{\text{rot}}^{\ddagger}$ have been discussed in [36]. We have performed RRKM calculations of the same type with decoupled $Q_{\text{vib}}^{\ddagger}$ and $Q_{\text{rot}}^{\ddagger}$; e.g. for a symmetric top

$$Q_{\text{rot}}^{\ddagger} = \sum_{K=0}^{J_{\text{rot}}} \sum_{J=K}^{J_{\text{rot}}} g(J, K) \cdot \exp\left(-\frac{\Delta E(J, K)}{kT}\right). \quad (32)$$

$\Delta E(J, K)$ is the difference between the lowest product energy level $V_{0,\infty}$ and the lowest energy level of the activated complex at the maximum in the effective potential for a rotational state (J, K) , with degeneracy $g(J, K)$. It should be noted that even for simple bond fissions the threshold energy of the reaction is not exactly equal to $V_{0,\infty}$ because of the residual zero point energy of the loosened bending vibrations of the activated complex. The difference between the threshold energy of the

reaction and $V_{0,\infty}$ is included in $Q^{\ddagger}$. If $Q_{\text{rot}}^{\ddagger}$ is evaluated with the complete sum Equation (32), one avoids an arbitrary assignment of moments of inertia to the activated complex and some other approximations usually made.

VI. Minimum Local Entropy Criteria

In the following section we shall compare the model of open adiabatic channels with calculations based on minimum local entropy criteria. To find the “critical configuration” (or the “activated complex”) of the reaction, Bunker and Pattengill [11] proposed to select “a value of the reaction coordinate such that the number of accessible internal states of the molecule is minimized”. This minimum entropy criterion is in the line of suggestions given in the early transition state theory [38] and apparently it corresponds to some dynamic bottle neck of the reaction [7, 13].

Originally the criterion was used [7, 8, 11] in the form

$$\frac{\partial \rho_{\text{vib}}(E, J, q)}{\partial q} = 0 \quad \text{at} \quad q = q^{\ddagger} \quad (33)$$

where q = reaction coordinate, $q^{\ddagger}$ = critical value of q . ρ is the number of quantum states in the range $(E, E + dE)$ and $(q, q + dq)$ and is obtained by convolution of the translational density of states [7, 8, 11] and the vibrational density of states (e.g. Equation (23)). Another possibility would be to exclude the translational states and use a minimum number of states criterion

$$\frac{\partial W(E, J, q)}{\partial q} = 0 \quad \text{at} \quad q = q^{\ddagger} \quad (34)$$

where W denotes the number of accessible quantum states of the non-critical coordinates (i.e. the molecular coordinates leaving out position and momentum in the reaction coordinate). For the calculation of both $\rho(q)$ and $W(q)$ neither the classical nor the semiclassical approximations are applicable in polyatomic systems. Only in triatomic systems the errors are small.

Once a critical configuration is chosen for a given energy E and rotational state J , the specific rate constant from transition state theory is given by [1–8]

$$k(E, J) = \frac{W^{\ddagger}}{h \cdot \rho(E, J)} \quad (35)$$

where $W^{\ddagger}$ denotes the number of accessible internal states of the molecule at the critical configuration. In simple bond fission reactions, the critical configurations $q^{\ddagger}$ in general do not correspond to the maxima of the potential surface, as shown in [11]. Therefore, there exists some ambiguity on which internal states should be included in W : one might either use $W^{\ddagger} = W(E - V_0(q^{\ddagger}))$ or $W^{\ddagger} = W(E - V_{0,\text{max}})$ (see Fig. 3). The former choice intuitively would correspond to more diabatic, the latter to more adiabatic motions at $q > q^{\ddagger}$. Neither possibility can be justified a priori. Therefore we have compared both possibilities with the present model.

In the following we compare the two local entropy criteria of Equations (33) and (34), in conjunction with the two possibilities to calculate $W^{\ddagger}$ at the activated complex, with the results of the open channel model. In order to avoid problems with

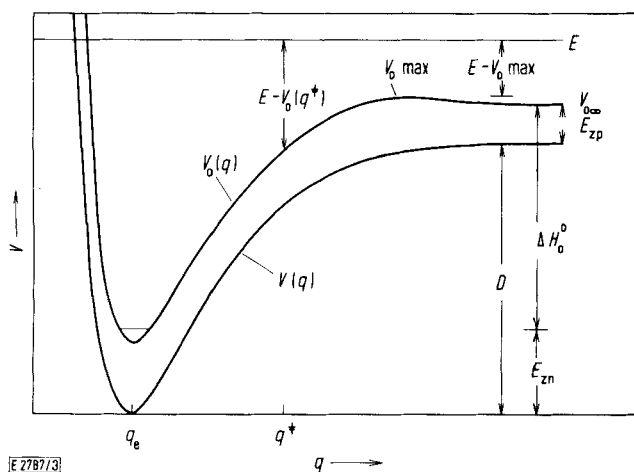


Fig. 3

Potential energy curves $V(q)$ (Equation (16)), and $V_0(q) = V(q) + \text{zero point energy of the non-critical coordinates}$. For the various energy parameters see text

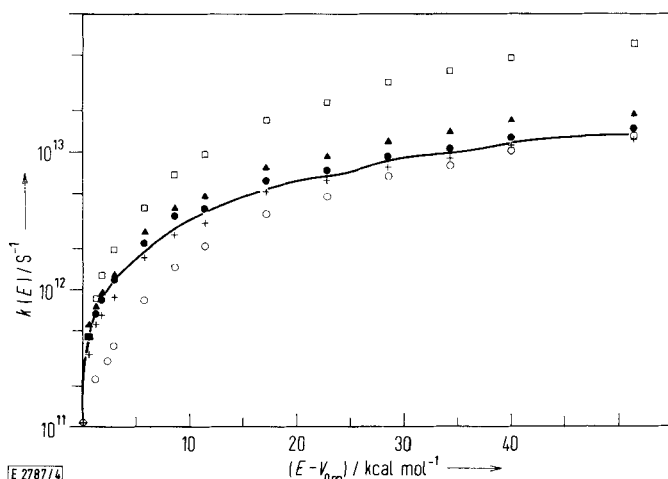


Fig. 4

Specific dissociation rate constants $k(E, J=0)$ for NO_2 (with $\alpha = 1.3 \text{ \AA}^{-1}$ see text); — open channel model (curve 1); + from $W^\ddagger = W(E - V_{0\text{max}})$ (curve 2); ● from $W^\ddagger = W(E - V_0(q^\ddagger))$ (curve 3); ▲ from $W^\ddagger = W(E - V_0(q^\ddagger))$ but with harmonic oscillator description for bending vibration (curve 4); □ from RRKM model, which gives an approximate fit to the experimental $k_\infty(T)$ (curve 5); ○ from RRKM model, which gives a fit to $k(E)$ open channel at high energies (curve 6); ◇ is the threshold rate constant for dissociation, common to all models

the vibration-rotation coupling, we have chosen the example of a nonrotating molecule. We consider the dissociation of NO_2 . All molecular parameters and details of the numerical calculations are given in the Appendix. In Fig. 4 the computed values of $k(E)$ are shown (with a Morse parameter $\beta = 3.35 \text{ \AA}^{-1}$ and a Pauling parameter $\alpha = 1.3 \text{ \AA}^{-1}$). Curve 1 gives the specific rate constant $k(E)$ for the open channel model, curve 2(+) gives $k(E)$ for the minimum number of states criterion Equation (34) and $W^\ddagger = W(E - V_{0\text{max}})$ and curve 3(●) gives the latter but for $W^\ddagger = W(E - V_0(q^\ddagger))$. As $W(E - V_0(q^\ddagger))$ is always larger than (or equal to) $W(E - V_{0\text{max}})$ it follows that at all energies $k(E)$ in curve 3 is larger

than in curve 2. As can be seen from the figure, the "correct" values of $k(E)$ (curve 1), obtained using the open channel model, lie between those of curves 2 and 3. This can be understood from the fact that not all states counted in $W(E - V_0(q^\ddagger))$ correspond to open channels; some channels counted arrive at q_{max} at energies $> E$ and are closed; $W(E - V_0(q^\ddagger))$ gives in general only an upper limit of the number of open channels. $W(E - V_{0\text{max}})$ generally gives a lower limit, because one does not count some channels whose energies decrease considerably with q . As the individual channel curves can have quite different forms, these statements need not be general and the following relation holds only approximately,

$$W(E - V_0(q^\ddagger)) \gtrsim W(E, J) \gtrsim W(E - V_{0\text{max}}). \quad (36)$$

However, in all examples we considered, Equation (36) was fulfilled, and $W(E - V_0(q^\ddagger))$ was generally a better approximation to the number of open channels $W(E, J)$ than $W(E - V_{0\text{max}})$. For large values of α and larger molecules, the difference between the three quantities (W) becomes more pronounced than shown in Fig. 4 for NO_2 dissociation and $W(E, J) \simeq W(E - V_0(q^\ddagger))$ at low energies, whereas $W(E, J) \simeq W(E - V_{0\text{max}})$ at high energies.

The two minimum local entropy criteria, Equations (33) and (34), give only slightly different results. As including the translational states leads to a highly discontinuous function $\rho(q)$, whose minimum has to be looked for, the choice of the minimum number of states criterion Equation (34) offers some advantages. In both cases, the strong energy dependence of the position of the critical surface found in [8] has been established.

The calculations illustrated in Fig. 4 show to what extent the minimum local entropy criteria are equivalent to a model of open adiabatic reaction channels (in the example the differences are within 10–30% but for larger molecules the differences are greater). As the open channel model involves the statistical postulate in a less restrictive way than the usual transition state theory with a localized activated complex, we consider it to be more realistic. The open channel model allows to follow individual reaction channels from the reactant to the products and to specify completely state-selected rate constants. As a link between quantum scattering theory and transition state theory, the present open channel model justifies, to some extent, the minimum local entropy criteria used in the most detailed versions of the transition state theory.

VII. Numerical Examples

We shall apply the model of open adiabatic channels to the examples $\text{NO}_2 \rightarrow \text{NO} + \text{O}$, $\text{CH}_4 \rightarrow \text{CH}_3 + \text{H}$, $\text{CD}_4 \rightarrow \text{CD}_3 + \text{D}$, $\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$, and $\text{C}_2\text{D}_6 \rightarrow 2\text{CD}_3$. All reactions are considered on the surface of the electronic ground state. The molecular constants used and details of the calculations are given in the Appendix. We shall compare with experiments and with RRKM calculations. Also the effect on $k(E)$ and k_∞ of the variation of the potential parameter α and the isotope effects will be studied.

Specific Rate Constants: Fig. 5 shows the specific rate constants $k(E, J=0)$ for NO_2 dissociation as a function of the Pauling parameter α . In Fig. 6 $k(E, J)$ is given for several J

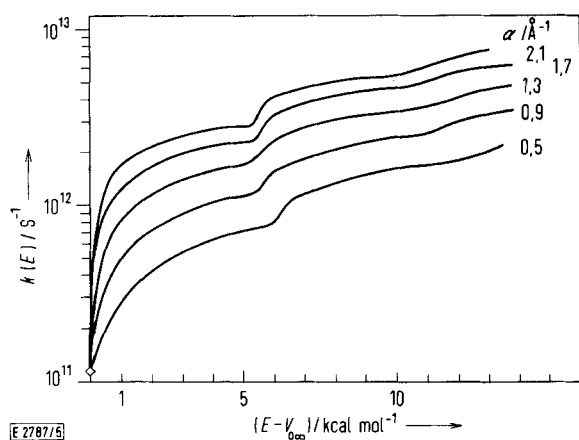


Fig. 5

Specific rate constants $k(E, J=0)$ for NO_2 as function of the parameter α

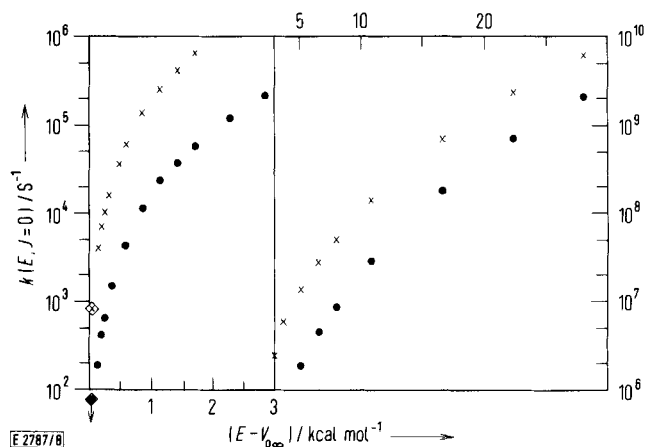


Fig. 8

Specific rate constants for ethane decomposition ($J=0$). Computed points: $\times$ C_2H_6 ; $\bullet$ C_2D_6 ; $\diamond$ rate constants at the thresholds for dissociation

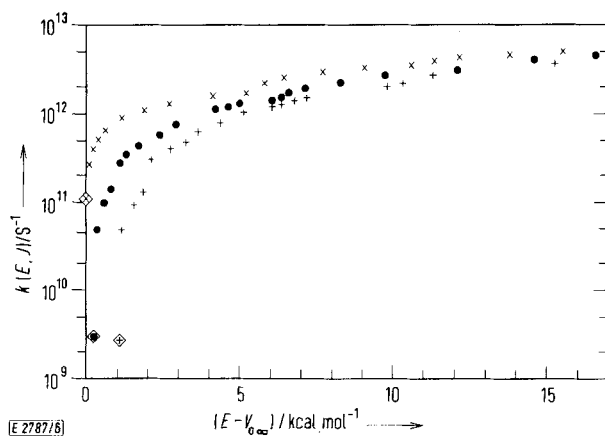


Fig. 6

Specific rate constants $k(E, J)$ for NO_2 dissociation ($\alpha = 1.3 \text{ \AA}^{-1}$). Computed points: $\times$ $k(E, J=0)$; $\bullet$ $k(E, J=30)$; $+$ $k(E, J=60)$; $\diamond$ rate constants at the individual thresholds for dissociation, neglecting tunneling through centrifugal barriers

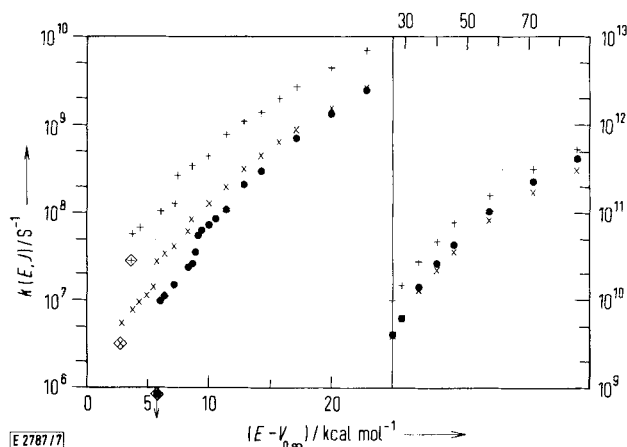


Fig. 7

Specific rate constants for methane decomposition. Computed points: $+$ $k(E, J=0)$, CH_4 ; $\times$ $k(E, J=40)$, CD_4 ; $\bullet$ $k(E, J=70)$, CD_4 ; $\diamond$ rate constants at the thresholds of dissociation

values. α was taken to be $1.3 \text{ \AA}^{-1}$ which gives a good fit to the experimental data on k_∞ (see below). With increasing J a decrease of $k(E, J)$ is observed. This is partly due to centrifugal barriers and partly to an increase in phase space accessible for the reactant because of "additional" rotational states entering $\rho(E, J)$ if one takes full account of the coupling of rotation and vibration. The latter makes a comparison between the minimum density of states calculation [8] and the present model difficult for $J > 0$. The results of these two models would become comparable, if one introduced an additional "internal rotation" corresponding to the quantum number K in the quasi-linear model for the molecule used in [8].

If one uses a harmonic oscillator model with constantly decreasing frequency for the bending vibration, one obtains curve 4 in Fig. 4. As is to be expected quite generally $k(E)$ values are higher, particularly for large energies, if compared to the more correct description as hindered rotation used in the present model (curves 1–3).

Curve 5 in Fig. 4 gives the result of an RRKM calculation with parameters chosen to give an approximate fit to the experimental k_∞ . The much steeper rise in $k(E)$ for the RRKM calculation leads to a higher value for the Arrhenius energy. Curve 6 in Fig. 4 is the result of an RRKM calculation chosen to agree with the $k(E)$ of the present model at high energies. This would lead to a much too low value for the thermal rate k_∞ . (The parameters used for curves 5 and 6 are given in the appendix.) The general trend is quite obvious: In an RRKM picture the present model would correspond to an activated complex, whose rigidity increases with increasing energy.

The rather pronounced isotope effect on the specific rate constants in both the C_2H_6 and CH_4 systems may be seen from Figs. 7 and 8.

Thermal Rate Constants k_∞ : The thermal high pressure rate constants k_∞ are experimentally accessible at high temperatures from dissociation studies and at low and intermediate temperatures, after conversion via the equilibrium constants, in recombination studies. In the following, k_∞ is represented in the form

$$k_{\infty} = A_{\infty}(T) \exp\left(-\frac{\Delta H_0^0}{RT}\right) \quad (37)$$

or, in the usual Arrhenius form

$$k_{\infty} \simeq A_{\infty}^* \exp\left(-\frac{E_{a\infty}}{RT}\right). \quad (38)$$

For the open channel model only the Pauling parameter α is treated as adjustable, although some uncertainty in the Morse parameter β remains. For the RRKM calculations the frequencies of the activated complex have been chosen to reasonably fit the experiments. The question is whether the models reproduce both the absolute values of the rate constants and the Arrhenius energies without introducing new parameters.

For NO_2 , the experiments [39, 40] (reevaluated with new reference rate constants [41] lead to a recombination rate of $k_{\text{rec}}(300 \text{ K}) = 1.7 \cdot 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and by combination with the dissociation rate constant at 1500 K [40] to a mean Arrhenius expression $k_{\infty} \simeq 10^{14.2} \exp(-71.2 \text{ kcal mol}^{-1}/RT) \text{ s}^{-1}$ between 300 K and 1500 K (both experimental values have been estimated to be accurate within a factor 2). Table 1

Table 1

$\alpha/\text{\AA}^{-1}$	$k_{\text{rec}}(300 \text{ K})/10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$	$E_{a\infty}/\text{kcal mol}^{-1}$
0.9	0.64	72.6
1.1	0.94	72.5
1.3	1.2	72.5
1.7	1.9	72.5
2.1	2.2	72.6
Exp.	1.7	≥ 71.2
"best" RRKM	1.0	73.3

Calculated recombination rate constants for NO_2 at 300 K and Arrhenius activation energies for dissociation between 300 and 1500 K as function of the parameter α (high pressure limit).

shows the calculated rate constant for recombination at 300 K and the Arrhenius energies for dissociation as a function of the parameter α together with the experimental quantities and the results of a RRKM calculation. A variation of α by a factor of more than 2 gives a variation of the absolute rate of less than a factor 4. The Arrhenius energy is essentially independent of α . A value of $\alpha = 1.3 \text{ \AA}^{-1}$ gives a reasonable fit to the experiments, but the calculated Arrhenius energy is high. This may be due to an insufficient representation of the potential surface by just one parameter (see below). The RRKM calculations leads to a still larger value of $E_{a\infty}$ as is to be expected from the $k(E)$ curve in Fig. 4.

For CH_4 , only dissociation data near 2200 K are available, which gave $A_{\infty}(2200 \text{ K}) = 10^{14.74} \text{ s}^{-1}$ [42] with $\Delta H_0^0 = 103.2 \text{ kcal mol}^{-1}$. The experimental value can only be reproduced with $\alpha \simeq 0.1 \text{ \AA}^{-1}$ (calculated $A_{\infty}(2200 \text{ K}) = 10^{14.79} \text{ s}^{-1}$) which corresponds to very high barriers to internal rotation for quite large CH_3-H bond extensions. With $\alpha \simeq 1.0 \text{ \AA}^{-1}$, $A_{\infty}(2200 \text{ K})$ would be about ten times the experimental value. A low α , on the other hand, introduces fairly large barriers to reaction, because of the residual zero point energies (see Fig. 7). This leads to a high value $E_{a\infty} \simeq 108 \text{ kcal mol}^{-1}$ whereas the experimental estimate at 2200 K is $E_{a\infty} \simeq 104 \pm 2 \text{ kcal mol}^{-1}$

[42]. Presently, we cannot decide whether these discrepancies are due to an incomplete approach of the high pressure limit in the experiments, or to our assumption $\gamma \simeq 1$ in Equation (2). This neglect of specific quantum effects may not be justified for a C-H bond cleavage. The predicted isotope effect $k_{\infty}(\text{CH}_4):k_{\infty}(\text{CD}_4)$ for dissociation at 2200 K is 1.5:1. The predicted rates for recombination at 300 K, $k_{\text{rec}}(\text{CH}_4) = 1.8 \cdot 10^9 \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $k_{\text{rec}}(\text{CD}_4) = 2.9 \cdot 10^9 \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ are quite low. As noted, the low α introduces a barrier for the recombination reaction, hence these values may be due to an artifact caused by incomplete knowledge of the potential surface. Again, an RRKM calculation leads to a still higher Arrhenius energy $E_{a\infty}(2200 \text{ K}) \simeq 109.5 \text{ kcal mol}^{-1}$.

For ethane decomposition and methyl radical recombination, there are fairly numerous experimental investigations [43–49]. With the results of the recombination measurements at 300 K [47–49] and the rate for dissociation at 900 K [44] and a reextrapolation to the high pressure limit as proposed in [43] one can represent the experimental data in this temperature range by $k_{\infty} \simeq 10^{17.1} \exp(-89.6 \text{ kcal mol}^{-1}/RT) \text{ s}^{-1}$. For our calculation we used a somewhat simplified procedure, correlating states of individual coordinates separately (see Appendix). This avoids the complex six-dimensional correlation problem (two doubly degenerate rocking vibrations, the torsion and the rotation about the figure axis becoming rotations of the methyl fragments). In the case of methane, we checked that this simplified method leads only to slight changes in k_{∞} when compared with the correct correlation. With a value of $\alpha = 1.0 \text{ \AA}^{-1}$ one obtains $k_{\infty} = 10^{17.2} \exp(-89.6 \text{ kcal mol}^{-1}/RT) \text{ s}^{-1}$ in good agreement with experiments in the range 300–900 K. The dissociation rate of C_2D_6 at 900 K ($k_{\infty} = 1.9 \cdot 10^{-5} \text{ s}^{-1}$) is predicted to be about 50% lower than for C_2H_6 ($k_{\infty} = 2.8 \cdot 10^{-5} \text{ s}^{-1}$). The predicted isotope effect for recombination at 300 K corresponds to $k_{\text{rec}}(\text{C}_2\text{H}_6) = 3.1 \cdot 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ and $k_{\text{rec}}(\text{C}_2\text{D}_6) = 2.4 \cdot 10^{13} \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ [50].

Table 2

Reference	Temperature	$\lg(A_{\infty}^*/\text{s}^{-1})$	$E_{a\infty}/\text{kcal mol}^{-1}$
Exp.	[44, 47, 43] 300–900 K	17.1	89.6
	[45] 850–930 K	16.0	86.0
Th.	[43] 900 K	18.1	90.6
	[12] 400–900 K	16.5	88.3
	this work with RRKM model 300–900 K	18.3	92.0
	this work 300–900 K	17.2	89.6

Arrhenius-parameters for ethane dissociation.

There have been several theoretical investigations on the $\text{C}_2\text{H}_6 \rightleftharpoons 2\text{CH}_3$ system. For convenience, we summarize some theoretical and experimental Arrhenius expressions in Table 2. Waage and Rabinovitch [43] recently emphasized an inconsistency between kinetic results, the thermochemistry and a detailed RRKM calculation. With $\Delta H_0^0 = 86.3 \text{ kcal mol}^{-1}$ they obtained $k_{\infty} = 10^{18.1} \exp(-90.6 \text{ kcal mol}^{-1}/RT) \text{ s}^{-1}$. If one corrects their result to a value of $\Delta H_0^0 = 87.76 \pm 0.2 \text{ kcal mol}^{-1}$ [55, 56], there is good agreement with our

RRKM calculation (see Table 2 and Appendix). The Arrhenius energy $E_{a\infty} \simeq 92 \text{ kcal mol}^{-1}$ obtained from the RRKM calculations in the range 300–900 K is definitely too high. This systematic defect is absent in our open channel model for the reasons already noted.

Recently [12] it has been claimed that all experimental data can well be represented with a fixed activated complex at $q^\ddagger = 3.6 \text{ \AA}$, chosen by the minimum density of states criterion. However, in this work an effective dissociation energy of $\simeq 84.9 \text{ kcal mol}^{-1}$ has been used, which is much lower than the thermodynamic value ΔH_0^0 . Hence this calculation cannot be adequate. If one corrects just the dissociation energy in [12], one obtains again a high Arrhenius energy of $E_{a\infty} \simeq 92 \text{ kcal mol}^{-1}$. We estimate, however, that a model which uses a minimum density (or number) of states criterion for every energy and rotational state separately and a correct threshold energy $\geq \Delta H_0^0$ should give results fairly close to our open channel calculation.

VIII. Conclusions

In the present model one tries to calculate, so far by simple interpolation, how the eigenstates of a reactant molecule during the reaction are transformed into the eigenstates of the products. The energy of a particular eigenstate as function of the reaction coordinate defines an "adiabatic potential curve" for the particular reaction channel. In the adiabatic approximation such a channel is "open" if the internal energy of the molecule is larger than the maximum of the adiabatic potential curve. By the additional statistical assumption, at a given energy all open adiabatic channels are populated with equal weight. The average rate constant for each channel follows from resonance scattering theory as $\gamma/h \cdot \rho(E)$, where γ is assumed to be nearly 1 for all channels above their individual threshold energies. In principle, the step function $\gamma(E)$ could be replaced by more detailed expressions calculated from resonance scattering theory. The question whether an experiment corresponds to a statistical situation has to be investigated separately. In any case, comparison with the statistical calculation of the present model will be useful for classifying experimental results.

The calculations illustrate which parts of the potential surface are particularly important for the considered processes and should be investigated by quantum-chemical methods. All deformation vibrations of the molecule which disappear during the reaction, at intermediate interfragment distances, are hindered rotations of the fragments relative to each others. It has been shown that the barriers for these hindered rotations, i.e. the minima and maxima of the potential energy as function of bond angles, at different interfragment distances, together with the potential energy in the reaction coordinate are particularly important parts of the potential surface for the considered processes. So far, we have used a simple one parameter interpolation between harmonic oscillator levels for the reactant states and free rotor levels for the product states. Although correct qualitatively, this approximation may be insufficient in more quantitative applications.

The present model provides an independent justification of the "minimum density of states" criterion proposed by

Bunker and Pattengill. Whenever the use of this criterion would seem simpler than the present open channel model, one may use Equation (36) to get an estimate of the uncertainties due to the different versions of the statistical postulate. It is essential for applications that the dynamical bottle-neck underlying this criterion is a strong function of the energy of the molecule. A considerable improvement over RRKM theory can be obtained only with these energy dependent positions of activated complexes.

With the present procedure one can easily compute rovibrational and translational energy distributions of dissociation products and bimolecular scattering cross sections if the reaction proceeds via an intermediate complex with a deep minimum.

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Appendix

Molecular Parameters

	$^{14}\text{N}^{16}\text{O}_2$	$^{12}\text{CH}_4$	$^{12}\text{CD}_4$	$^{12}\text{C}_2\text{H}_6$	$^{12}\text{C}_2\text{D}_6$
A_0/cm^{-1}	8.00213	5.2412	2.66304	2.671	1.342
B_0/cm^{-1}	0.433686	5.2412	2.66304	0.66310	0.45973
C_0/cm^{-1}	0.41044	5.2412	2.66304	0.66310	0.45973
$\Delta H_0^0/\text{cm}^{-1}$	25134.	36110.	36900.	30708.	31401.
$q_e/\text{\AA}$	1.1934	1.091	1.091	1.53	1.53
D/cm^{-1}	26026.	39220.	39220.	33859.	33859.
$\beta/\text{\AA}^{-1}$	3.35	3.0	3.0	1.8	1.8
$\alpha/\text{\AA}^{-1}$	1.3	0.1	0.1	1.0	1.0
Ref.	[51, 52, 53]	[51, 54, 55]		[55, 56, 57, 58]	

	Species	ν/cm^{-1} $^{14}\text{N}^{16}\text{O}_2$	Description *)
ν_1	A_1	1319.7	stretch (1)
ν_2	A_1	749.8	bend (1)
ν_3	B_2	1617.75	stretch (1)
$\nu_{\text{NO}} = 1876 \text{ cm}^{-1}$ $B_{\text{NO}} = 1.696 \text{ cm}^{-1}$			
Ref. [52]			

	Species	ν/cm^{-1} $^{12}\text{CH}_4$	$^{12}\text{CD}_4$	Description *)
ν_1	A_1	2916.5	2107.9	stretch (1)
ν_2	E	1533.6	1092.2	bend (2)
ν_3	F_2	3019.49	2258.6	stretch (3)
ν_4	F_2	1305.9	996.0	bend (3)

Ref. [51, 54]

	Species	ν/cm^{-1} $^{12}\text{CH}_3$	$^{12}\text{CD}_3$	Description *)
ν_1	A'_1	3044	2153	stretch (1)
ν_2	A'_2	580	447	out of plane bend (1)
ν_3	E'	3162	2381	stretch (2)
ν_4	E'	1396	1026	in plane bend (2)
	A_0/cm^{-1}	4.789	2.3965	$B_0 = C_0 = 2A_0$

Ref. [59, 60, 61]

	Species	ν/cm^{-1}		Description *)
		C_2H_6	C_2D_6	
ν_1	A_{1g}	2954	2083	CH_3 -stretch (1)
ν_2	A_{1g}	1388	1155	CH_3 deformation (1)
ν_3	A_{1g}	995	843	C—C stretch (1)
ν_4	A_{1u}	289 (288)**)	208 (210)**)	Torsion (1)
ν_5	A_{2u}	2896	2087	CH_3 -stretch (1)
ν_6	A_{2u}	1379	1077	CH_3 deformation (1)
ν_7	E_u	2985	2234	CH_3 -stretch (2)
ν_8	E_u	1472	1081	CH_3 deformation (2)
ν_9	E_u	821	594	CH_3 -rock (2)
ν_{10}	E_g	2969	2225	CH_3 -stretch (2)
ν_{11}	E_g	1469	1041	CH_3 deformation (2)
ν_{12}	E_g	1206	970	CH_3 -rock (2)

Ref. [58, 62]

*) degeneracies in parenthesis.

**) values computed with a cosine potential with $V_3 = 1024 \text{ cm}^{-1}$.
 ν = fundamentals

Reactant — Product Correlations

 NO_2 : $\nu_1 \leftrightarrow q$; $\nu_3 \cong \nu_{\text{bn}}$; $\nu_2 \cong \nu_{\text{n}}$; $\nu_{\text{NO}} \cong \nu_{\text{p}}$ in Equation (14) CH_4, CD_4 : $\nu_1 \leftrightarrow q$

methane methyl

 $\nu_3 \dots \nu_3, \nu_1$ $\nu_4 \dots \nu_2, \nu_4$ $(\nu_2, \tau) \dots w \dots (j, K, l)$ $\text{C}_2\text{H}_6, \text{C}_2\text{D}_6$: $\nu_3 \leftrightarrow q$

ethane methyl

 $\nu_1, \nu_5, \nu_7, \nu_{10} \dots \nu_1, \nu_3$ $\nu_2, \nu_6 \dots \nu_2$ $\nu_8, \nu_{11} \dots \nu_4$ $J \dots l$ $\nu_9, \nu_{12} \dots j_1, j_2$ $\nu_4, K \dots K_1, K_2$

This simplified correlation is not supposed to give accurate $k(E, J)$, but rather good k_∞ as has been tested with CH_4 . Eigenvalues: stretching and bending vibrations

 $E_{\text{sb}}(q) \cong \nu_{\text{sb}} \cdot \{\nu_{\text{sb}}(\text{methyl})$ $+ [\nu_{\text{sb}}(\text{ethane}) - \nu_{\text{sb}}(\text{methyl})] \exp(-\alpha(q - q_c))\}$ Torsion: $E_4(q) \cong (c/2) \cdot (3n + 2)^2$ or $E_4(q) \cong (c/2) \cdot (3n + 2)^2$ $+ [n \cdot \nu_4 - (c/2)(3n + 2)^2] \cdot \exp(-\alpha(q - q_c))$

whichever is larger, with n = quantum number of torsion and c = rotational constant of methyl fragment.

Rocking vibrations:

 $E_r(q) \cong A_0 \cdot m(m + 1)$ $+ [m \cdot \nu_9 - A_0 m(m + 1)] \exp(-\alpha(q - q_c))$

m = quantum number of rocking vibration. Internal rotation about the C—C bond leads to a splitting of the torsional levels, which are threefold degenerate. Also, for the rocking

vibrations, there is a fourfold inversion type degeneracy, which is lifted by the possibility of completely free rotation of the methyl fragments for large amplitudes [32, 33]. The formulae given for $E_4(q)$ and $E_r(q)$ correspond to mean energies for several levels which is a sufficient approximation for the present, rough computations.

Rotation: With the assumption of constant bond angles, rotational constants as function of extension of the reaction coordinate follow from the inertial tensor, e.g. for NO_2

$$\frac{B(q) + C(q)}{2} = 8.428 \cdot [I_{zz}^{-1} + (I_{zz}/2 + \sqrt{I_{zz}^2/4 + I_{xy}^2 - I_{yy} \cdot I_{xx}})^{-1}] \text{ cm}^{-1} \cdot \text{amu} \cdot \text{\AA}^2$$

with: $I_{xx} = 7.664 \text{ amu} \cdot \text{\AA}^2$; $I_{yy} = I_{zz} - I_{xx}$; $I_{zz}/\text{amu} \cdot \text{\AA}^2 = 10.435 \cdot (q/\text{\AA})^2 + 9.242 (q/\text{\AA}) + 14.862$; $I_{xy}/\text{amu} \cdot \text{\AA}^2 = 7.427 + 4.786 (q/\text{\AA})$.

The equilibrium constants necessary for computing recombination rates have been calculated with the molecular data given, in the usual harmonic oscillator, rigid rotor approximation. The energy levels used for torsion of C_2H_6 and C_2D_6 are eigenvalues of the Mathieu equation [63] with a common barrier of $V_3 = 1024 \text{ cm}^{-1}$. (We are indebted to Prof. A. Bauder, ETH, Zürich, for sending us a program for computation of these eigenvalues).

Parameters of the RRKM models: The $\Delta E(J, K)$ of Equation (32) have been calculated with rotational constants as function of q but with fixed bond angles. The activated complex frequencies in cm^{-1} , which give an approximate fit to the experimental high pressure rate constants, are shown in the following with degeneracies in parenthesis.

 NO_2 : 1883 (1); 53 (1); $n^\# = 2$ CH_4 : 3100 (3); 1350 (2); 1088 (1); 1073 (2); $n^\# = 4$

C_2H_6 : 2950 (6); 1470 (4); 660 (2); 120 (2); 82 (2); Torsion = free rotor; $n^\# = 1$ (all torsional sublevels having been counted for Q).

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